

Untitled

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1 SFT RepNoise Results

Table 1: Experiment settings

Method	Setting	Value
SFT	Model	meta-llama/Llama-2-7b-hf
	Attack data mix	harmful: PKU-Alignment/BeaverTails_dangerous; benign: sst2 (data/sst2.json)
	Sample size	5000
	Test size	1000
	Harmful ratios	1%, 5%, 10%
	Learning rates	0.00001, 0.00005, 0.0001
	Epochs	5, 10, 20
	Train batch size	5
	Eval batch size	5
	Grad accum. steps	1
	Weight decay	0.1
	Warmup ratio	0.1
	Scheduler	constant
	Safety eval datasets	beavertails, advbench
RepNoise	Utility eval tasks	sst2, gsm8k, agnews
	Model	meta-llama/Llama-2-7b-hf
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Table 2: agnews (higher is better) at harmful ratio 1%

Learning rate	Method	Epoch 5	Epoch 10	Epoch 20
0.00001	SFT	59.7	60.8	38.1
	RepNoise	49.1	50.8	–
0.00005	SFT	59	56.3	63.6
	RepNoise	–	–	–
0.0001	SFT	61	58	54
	RepNoise	46.3	–	–

Table 3: agnews (higher is better) at harmful ratio 5%

Learning rate	Method	Epoch 5	Epoch 10	Epoch 20
0.00001	SFT	70.1	62.1	52.8
	RepNoise	–	–	–
0.00005	SFT	63.3	60.3	58.4
	RepNoise	–	–	–
0.0001	SFT	62.5	62.4	56.5
	RepNoise	–	–	–

Table 4: agnews (higher is better) at harmful ratio 10%

Learning rate	Method	Epoch 5	Epoch 10	Epoch 20
0.00001	SFT	70	65	41.9
	RepNoise	53.7	54.9	–
0.00005	SFT	63.3	43.2	38.4
	RepNoise	–	–	–
0.0001	SFT	56.3	50.8	53.6
	RepNoise	67.8	–	–

Table 5: gsm8k (higher is better) at harmful ratio 1%

Learning rate	Method	Epoch 5	Epoch 10	Epoch 20
0.00001	SFT	4.6	4.9	4.3
	RepNoise	4.9	5	4.1
0.00005	SFT	5.3	4.3	3.7
	RepNoise	–	–	–
0.0001	SFT	4.2	3.6	3.7
	RepNoise	4.7	4.4	4.8

Table 6: gsm8k (higher is better) at harmful ratio 5%

Learning rate	Method	Epoch 5	Epoch 10	Epoch 20
0.00001	SFT	5.3	5.5	5.5
	RepNoise	5.1	4.7	–
0.00005	SFT	5.6	5.3	3.6
	RepNoise	–	–	–
0.0001	SFT	5.4	4.5	4.1
	RepNoise	5	4.7	–

Table 7: gsm8k (higher is better) at harmful ratio 10%

Learning rate	Method	Epoch 5	Epoch 10	Epoch 20
0.00001	SFT	5.4	4.6	4.5
	RepNoise	4.2	4.6	4.5
0.00005	SFT	5.1	4.5	3.2
	RepNoise	–	–	–
0.0001	SFT	6	4.2	3.3
	RepNoise	4.7	3.6	3.3

Table 8: sst2 (higher is better) at harmful ratio 1%

Learning rate	Method	Epoch 5	Epoch 10	Epoch 20
0.00001	SFT	95.53	96.1	94.15
	RepNoise	95.64	95.99	95.64
0.00005	SFT	94.61	95.76	95.41
	RepNoise	–	–	–
0.0001	SFT	95.07	95.87	95.41
	RepNoise	95.53	95.3	96.1

Table 9: sst2 (higher is better) at harmful ratio 5%

Learning rate	Method	Epoch 5	Epoch 10	Epoch 20
0.00001	SFT	95.41	95.3	95.18
	RepNoise	95.3	96.1	–
0.00005	SFT	95.64	94.61	94.15
	RepNoise	–	–	–
0.0001	SFT	95.41	94.84	94.5
	RepNoise	95.53	95.3	–

Table 10: sst2 (higher is better) at harmful ratio 10%

Learning rate	Method	Epoch 5	Epoch 10	Epoch 20
0.00001	SFT	95.18	95.99	94.5
	RepNoise	94.38	95.87	95.53
0.00005	SFT	94.95	95.07	94.72
	RepNoise	–	–	–
0.0001	SFT	94.84	91.17	91.51
	RepNoise	95.64	94.84	95.76

Table 11: advbench (lower is better) at harmful ratio 1%

Learning rate	Method	Epoch 5	Epoch 10	Epoch 20
0.00001	SFT	60.38	70.96	79.04
	RepNoise	63.46	70.19	73.27
0.00005	SFT	78.27	70.38	71.73
	RepNoise	–	–	–
0.0001	SFT	74.81	73.08	70
	RepNoise	66.73	66.35	64.81

Table 12: advbench (lower is better) at harmful ratio 5%

Learning rate	Method	Epoch 5	Epoch 10	Epoch 20
0.00001	SFT	76.35	83.65	87.88
	RepNoise	76.54	84.42	–
0.00005	SFT	88.08	83.46	79.81
	RepNoise	–	–	–
0.0001	SFT	86.15	85	81.35
	RepNoise	85.38	87.12	–

Table 13: advbench (lower is better) at harmful ratio 10%

Learning rate	Method	Epoch 5	Epoch 10	Epoch 20
0.00001	SFT	83.85	88.27	88.46
	RepNoise	87.69	85.77	86.92
0.00005	SFT	86.54	82.69	87.88
	RepNoise	–	–	–
0.0001	SFT	84.62	85	84.81
	RepNoise	85.58	82.88	85.38

Table 14: beavertails (lower is better) at harmful ratio 1%

Learning rate	Method	Epoch 5	Epoch 10	Epoch 20
0.00001	SFT	54.6	62.1	68.7
	RepNoise	56	61.2	67.8
0.00005	SFT	67.6	68.3	71.2
	RepNoise	–	–	–
0.0001	SFT	70.9	69.7	70
	RepNoise	65.4	64.3	67.1

Table 15: beavertails (lower is better) at harmful ratio 5%

Learning rate	Method	Epoch 5	Epoch 10	Epoch 20
0.00001	SFT	66.5	74.1	74.6
	RepNoise	66.9	75.6	–
0.00005	SFT	76	75.7	76.9
	RepNoise	–	–	–
0.0001	SFT	77.1	76.9	77.1
	RepNoise	77.7	80	–

Table 16: beavertails (lower is better) at harmful ratio 10%

Learning rate	Method	Epoch 5	Epoch 10	Epoch 20
0.00001	SFT	72.6	76.8	77.8
	RepNoise	76.8	78.9	78.3
0.00005	SFT	77.2	78.9	79.4
	RepNoise	–	–	–
0.0001	SFT	76.4	77.6	79
	RepNoise	78.1	79.5	79